

# TD7: Ingeniería de Datos

Privacidad y Compliance

# Introducción

O'REILLY®

# Data Quality Fundamentals

A Practitioner's Guide to Building  
Trustworthy Data Pipelines



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# Introducción

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- Los datos recorren un camino desde que adquirimos hasta que los descartamos.
- El **Data Governance Lead** fija la estrategia de la organización en cuanto a la gobernanza de datos.
- A lo largo de este proceso, nuestra organización deberá cumplir con **normativas** relacionadas a la **protección de los datos personales**.
- El rol responsable de esta tarea a diario, y más en general de las tareas diarias de gobernanza de datos en una organización, es el **Data Steward**.

# Catálogos

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- Los catálogos (**data catalogs**) son metadatos curados relativos a **qué** datos tiene una organización, **quién** los usa, y en **donde** están.
- Representan una herramienta valiosa para la **gobernanza de datos**.
- Ayudan a saber si estamos cumpliendo con la normativa de cada país.
- Existen herramientas automatizadas de mantenimiento de catálogos.

# Designing a Data Catalog

Analogous to a physical library catalog, data catalogs serve as an inventory of metadata and give investors the information necessary to evaluate data accessibility, health, and location. Solutions that not only keep tabs on your data but also integrate with machine learning and automation to make data more discoverable, collaborative, and now, in compliance with organizational, industry-wide, or even government regulations.

Data catalogs are designed to answer questions such as:

- Where should I look for my data?
- Does this data matter?
- What does this data represent?
- Is this data relevant and important?
- How can I use this data?

Next-generation catalogs will have the capabilities to learn, understand, and infer the data, enabling users to leverage its insights in a self-service manner. Most importantly, data catalogs will support automated **data discovery** and active metadata.

Data discovery replaces the need for a data catalog by providing a domain-specific, dynamic understanding of your data based on how it's being ingested, stored, aggregated, and used by a set of specific consumers. As with a data catalog, governance standards and tooling are federated across these domains (allowing for greater accessibility and **interoperability**), but unlike a data catalog, data discovery surfaces a real-time understanding of the data's current state as opposed to its ideal or "cataloged" state.

# Data discovery defined

Data discovery is the process of extracting meaningful insights and patterns from a vast ocean of data.

Imagine being able to see trends, identify opportunities, and mitigate risks in real time. It's not about collecting more data; it's about making sense of the data you already have and transforming it into knowledge that can guide your business strategy.



## Engineering

Overview

Backend

Culture

Data / ML

Mobile

Uber AI

Web

Research

Uber

Schedule rides  
in advance

Reserve a ride

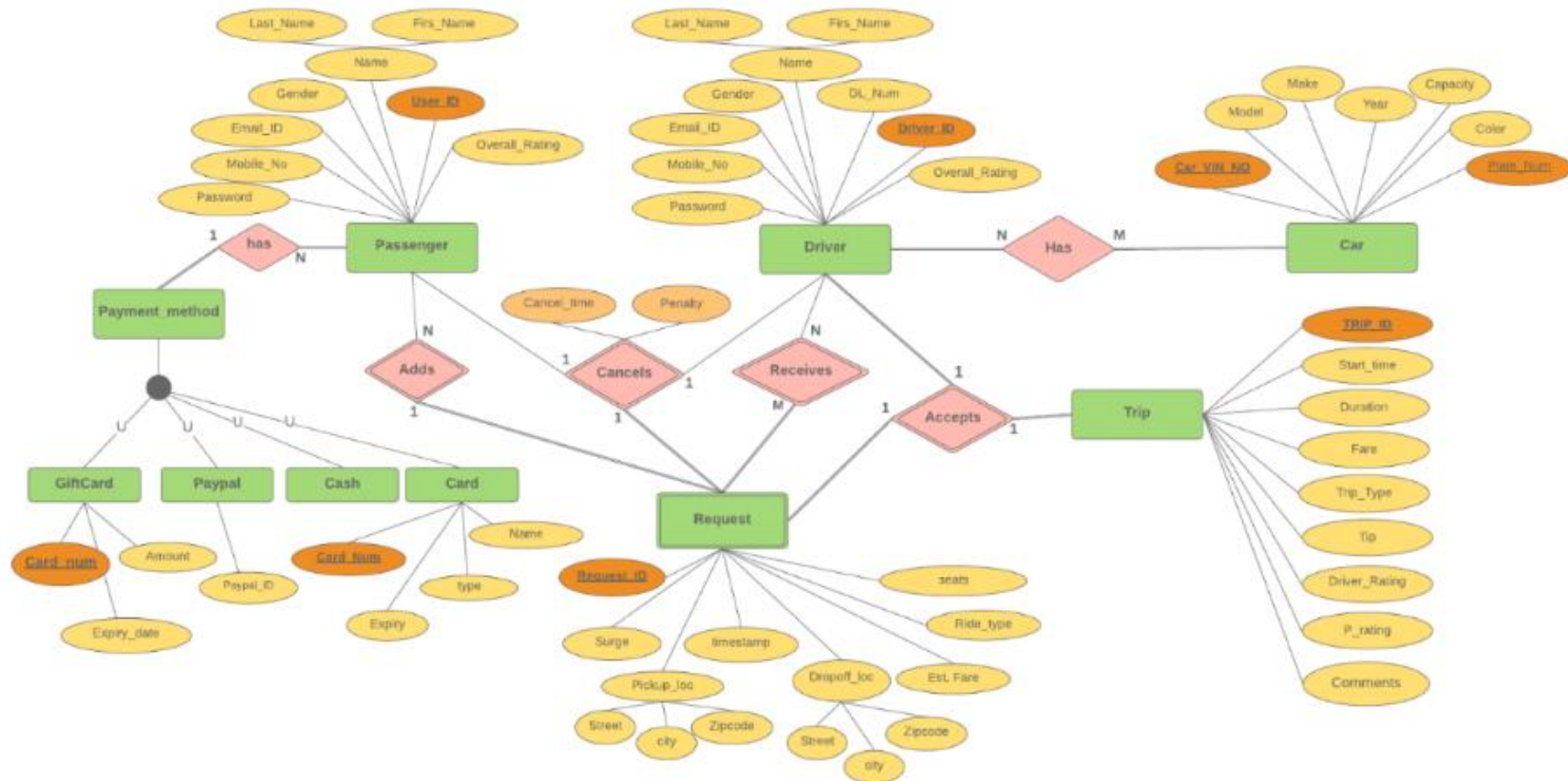
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Data / ML, Engineering

# Turning Metadata Into Insights with Databook

[Ejemplo: Uber Databook](#)

# ER Diagram for UBER





# Databook

Every day in over **10,000 cities** around the world, millions of people rely on Uber to travel, order food, and ship cargo. Our apps and services are available in over **69 countries** and run **24 hours a day**. At our global scale, these activities generate large amounts of logging & operational data that runs through our systems in real-time. This includes information about consumer demand, driver-partner availability, and other operational tasks such as payments, notifications, and more. Operating a complex marketplace such as ours requires our engineers, data scientists, data analysts, and operations managers to take real-time business decisions based on trends observed on our platform.



# Databook

- To help our teams easily discover and understand this data better, we built Databook. This is Uber's in-house platform which surfaces and manages the metadata related to various data entities such as datasets, internal dashboards, business metrics, and more.
- As Uber expanded, our systems became more technically complex and the breadth of our data systems grew exponentially. Similarly, our teams' needs became more complex as well, as they now relied even more on datasets, dashboards, and business metrics to make business decisions on a global scale.
- Databook was unable to scale to meet these new needs, prompting us to rethink our assumptions and rebuild our tools to better support our users and their evolving needs.

# Arriving at data democratization through data discovery

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Scaling data fluency and data discoverability go hand in hand, as analysts need to have easy access to data to make data-informed decisions.

## Data reactive

No one accesses or uses data in their daily work. Your company rarely reports on or presents data

## Data scaling

Very few people have the skills and access they need to analyze, report, and present data confidently

## Data progressive

Every team has at least one data fluent employee who can analyze, report, and present their data, regardless of role

## Data fluent

Everyone knows how to access the data they need to do their job (this doesn't mean that everyone needs to code)

# Aspectos legales de la protección de datos

# Información sensible - PII

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- Las empresas de la economía digital manejan bases de datos con información sensible de los ciudadanos.
- La mayor parte de los países de la OMC han promovido normativas para la protección de los ciudadanos en lo que respecta al almacenamiento y uso de sus datos personales.
- En este contexto, se denomina **PII (personally identifiable information)** a aquella información que permitiría identificar en forma directa a un individuo (ej., nombre, dirección, CUIT, teléfono, e-mail).

# Prioritizing Data Governance and Compliance

Perhaps there's no topic in all of the larger data quality discussion that draws as much **confusion and ire** as data governance, in other words, the management of data across and beyond an organization. Data governance is top of mind for many data leaders, particularly in light of GDPR, CCPA, IPOs, COVID-19, and any number of other acronyms that speak to the increasing importance of compliance and privacy when it comes to managing your company's data.



# Prioritizing Data Governance and Compliance

Traditionally, data governance refers to the process of maintaining the availability, usability, provenance, and security of data, and, as a data leader once told us, is the “keep your CFO out of jail card”. Still, Gartner suggests that more than 80% of data governance initiatives will fail in 2022.

# Prioritizing Data Governance and Compliance

In our opinion, data governance gets a bad reputation, primarily because traditional approaches fail to scale with the needs of cloud-based data stacks. Over the past several years, data catalogs have emerged as a powerful tool for data governance. As companies digitize and their data operations democratize, it's important for all elements of the data stack, from warehouses to business intelligence platforms, and now, catalogs, to participate in compliance best practices.

# Prioritizing Data Governance and Compliance

Data catalogs can be used to store metadata that gives stakeholders a better understanding of a specific source's lineage, thereby instilling greater trust in the data itself. Additionally, data catalogs make it easy to keep track of where **personally identifiable information (PII)** can both be housed and sprawl downstream, as well as who in the organization has the permission to access it across the pipeline.

# Prioritizing Data Governance and Compliance

There are three major types of automated data catalogs on the market today:

## In-house

Some B2C companies –Airbnbs, Netflixs, Ubers of the world– build their own data catalogs to ensure data compliance with state, country, and even economic union level regulations.

# Prioritizing Data Governance and Compliance

There are three major types of automated data catalogs on the market today:

## Third-party

Traditionally, data catalogs were managed manually and governed in **silos**, often requiring duplicated work between different analysts and data science teams. Now, there are a whole host of ML-powered data catalogs on the market that lend themselves to distributed governance, many with pay-for-play workflow and repository-oriented compliance management integrations. Some cloud providers, like Google, AWS, and Azure, also offer data governance tooling integration at an additional cost.

# Prioritizing Data Governance and Compliance

There are three major types of automated data catalogs on the market today:

## Open source

In 2018, Lyft became an industry leader by making their data discovery and metadata engine, Amundsen, named after the famed Antarctic explorer, open source. Other open source tools, such as Apache Atlas, Magda, and CKAN, provide similar functionalities

# Antecedentes de la protección de datos

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- “El derecho a la intimidad”, 1890, S. Warren, L. Brandeis, Harvard Law Review.

# Antecedentes de la protección de datos

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Recent inventions and business methods call attention to the next step which must be taken for the protection of the person, and for securing to the individual what Judge Cooley calls the right "to be let alone." Instantaneous photographs and newspaper enterprise have invaded the sacred precincts of private and domestic life; and numerous mechanical devices threaten to make good the prediction that "what is whispered in the closet shall be proclaimed from the house-tops."



# Antecedentes de la protección de datos

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The truth of the matter published does not afford a defense.

The absence of "malice" in the publisher does not afford a defense.

# Antecedentes de la protección de datos

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- [Declaración Universal de Derechos Humanos \(1948\)](#): Art. 12, toda personas tienen derecho a la protección de ley contra injerencias arbitrarias en su vida privada, su familia, su domicilio o su correspondencia.

# Antecedentes de la protección de datos

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- Privacy Act (Estados Unidos, 1974): Regula la recolección, mantenimiento, uso y diseminación de información sobre los individuos por parte de los agencias federales.

# Antecedentes de la protección de datos

Privacy Act:

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- Maintain Only Relevant and Necessary Information
- Collect Information, to the Greatest Extent Practicable, Directly from the Individual
- Inform Individuals when Asking to Collect Information
- Publish System of Records Notice

the name and location of the system

each routine use of the records contained in the system.

the policies and practices of the agency regarding storage, retrievability, access controls, retention, and disposal of the records;

the title and business address of the agency official who is responsible for the system of records;

# En Argentina



- Ley de Protección de los Datos Personales: (Ley 25326/2000). Los ciudadanos pueden, en relación a los datos personales guardados en archivos, registros, bancos de datos públicos o privados, solicitar:
  - que sus datos personales no sean utilizados ni registrados sin su consentimiento;
  - que les den información sobre qué datos personales están registrados en bancos de datos públicos o privados;
  - que sus datos sean corregidos o actualizados;
  - que sean suprimidos, en los casos en que corresponda;
  - que sean guardados confidencialmente;
  - iniciar una acción judicial para conocer sus datos o exigir su rectificación, supresión, confidencialidad o actualización.



Argentina.gob.ar

Buscá en el sitio



Mi Argentina

Jefatura de Gabinete de Ministros / Vicejefatura de Gabinete Ejecutiva /

# AAIP

Monitoreamos el desempeño de la transparencia en la gestión pública,  
garantizamos el acceso a la información y la protección de datos personales.



Acceso a la  
Información



Protección de  
Datos Personales



Evaluación de la  
Transparencia



Consejo Federal  
para la  
Transparencia



Trámites y servicios



Normativa

## Servicios destacados



### **Conocé tu derecho de acceso a la información**

Te explicamos la importancia de ejercerlo y quiénes están obligados a darte información.



### **Solicitá información**

Te explicamos cómo hacerlo en línea, por correo postal o de manera personal.



### **¿Te negaron información? Reclamá**

En la Agencia recibimos tu reclamo y nos encargamos de que hagas efectivo tu derecho.



### **Consultá los datos abiertos**

En los portales de los organismos nacionales podés encontrar información disponible en formatos abiertos.

## En la Unión Europea



- La Unión Europea posee desde 2016 un “Reglamento General de Protección de Datos” (GDPR)

## En Estados Unidos



- En 2018 se introdujo el California Consumer Privacy Act [CCPA]



Data leaks are costly and painful, as are regulatory fines.

*Security and compliance*

# Recolección de información sensible

# Recolección de información sensible

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## ¿Qué son los datos personales?

Cualquier tipo de información: datos de identidad, de domicilio, de deudas.

Imagen en videos de sistema vigilancia

Datos biométricos son datos personales

No puede haber registros de datos referidos al origen racial o étnico, opiniones políticas, convicciones religiosas, filosóficas o morales, afiliación sindical o que se relacionen con la salud o la vida sexual

# Recolección de información sensible

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- Registros en sitios web
- Cookies
- Spyware
- RFIDs, geolocalización

# Data literacy

# Data literacy (alfabetización de datos)

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- La sociedad digital requiere de los ciudadanos la habilidad de manejar datos correctamente.
- El desarrollo de estas habilidades no solo es una herramienta de inserción para los ciudadanos, sino que desde el punto de vista organizacional disminuye las posibilidades de *data leakages* y favorece el trabajo conjunto en pos de la gobernanza de datos.

# Data literacy

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## Swampification

Often, you'll have to rely on a skilled data engineer or data scientist for knowledge about where certain data resides, who its stakeholders are, and how it might be expected to change. Lean too heavily on this and your data “swampifies”, meaning no one can get any work done because data literacy comes at a high learning curve.

# Data literacy

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## Increasing Data Literacy

We've discussed the “what” and “who” of democratizing data quality, but “how” do you actually build a culture of data quality? For many organizations, it all starts with data literacy, in other words, the ability to read, write, and communicate about data in a way that drives value and impact for the organization. After all, how can you understand the value of data quality if you don't even understand the value of data? Or, for that matter, know how to use it?



# Data literacy

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## A good data literacy strategy

will gain top-down buy-in and bottoms-up adoption by making data more accessible and easier to work with, leveraging self-service tooling and education for less technical team members. The way to make these data initiatives successful beyond the boundaries of the data function and impactful for the broader organization, then, is to meet data stakeholders where they are.

# Data literacy

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- Por eso, a nivel gubernamental se están planteando objetivos para que los ciudadanos alcancen en gran medida la alfabetización de datos. [Ejemplo en la UE](#)



# European data

data.europa.eu

The official portal for European data

Home

Data ▾

EU Open Data Days

Academy

Community ▾

Publications ▾

Documentation [↗](#)

[Home](#) > [Publications](#) > [Data stories](#) > Digital literacy in the EU: An overview

06 December 2023

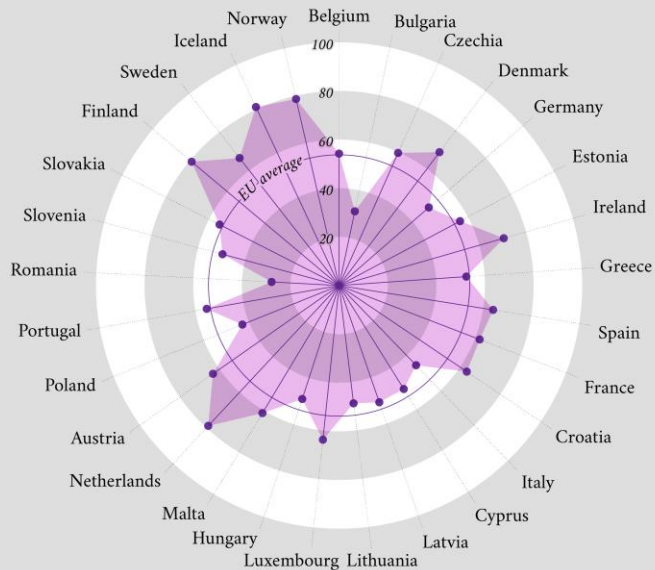
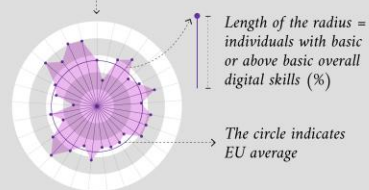
## Digital literacy in the EU: An overview

Measuring and improving digital proficiency through data

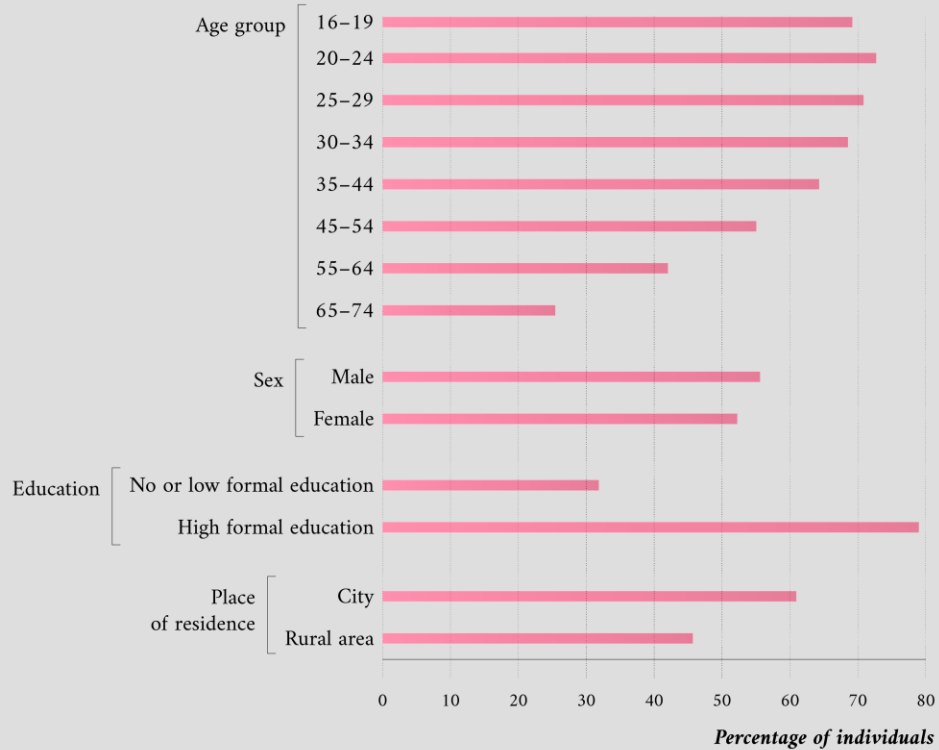
## INDIVIDUALS WITH BASIC OR ABOVE BASIC OVERALL DIGITAL SKILLS BY COUNTRY, 2021

### LEGEND

Each radius is a country



## INDIVIDUALS WITH BASIC OR ABOVE BASIC OVERALL DIGITAL SKILLS BY SOCIOECONOMIC STATUS, 2021



Source: Eurostat.

# Data literacy

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- [The Digital Competence Framework for Citizens](#)



DigComp 2.2

# The Digital Competence Framework for Citizens

*With new examples of knowledge, skills and attitudes*

Riina Vuorikari  
Stefano Kluzer  
Yves Punie



#### DIMENSION 1 • COMPETENCE AREA

### 1. INFORMATION AND DATA LITERACY

#### DIMENSION 2 • COMPETENCE

### 1.1 BROWSING, SEARCHING AND FILTERING DATA, INFORMATION AND DIGITAL CONTENT

To articulate information needs, to search for data, information and content in digital environments, to access them and to navigate between them. To create and update personal search strategies.

#### DIMENSION 3 • PROFICIENCY LEVEL

|                    |   |                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------|---|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FOUNDATION         | 1 | At basic level and with guidance, I can:                                                            | <ul style="list-style-type: none"> <li>identify my information needs, find data, information and content through a simple search in digital environments,</li> <li>find how to access these data, information and content and navigate between them,</li> <li>identify simple personal search strategies.</li> </ul>                                                                             |
|                    | 2 | At basic level and with autonomy and appropriate guidance where needed, I can:                      | <ul style="list-style-type: none"> <li>identify my information needs,</li> <li>find data, information and content through a simple search in digital environments,</li> <li>find how to access these data, information and content and navigate between them,</li> <li>identify simple personal search strategies.</li> </ul>                                                                    |
| INTERMEDIATE       | 3 | On my own and solving straightforward problems, I can:                                              | <ul style="list-style-type: none"> <li>explain my information needs,</li> <li>perform well-defined and routine searches to find data, information and content in digital environments,</li> <li>explain how to access them and navigate between them,</li> <li>explain well-defined and routine personal search strategies.</li> </ul>                                                           |
|                    | 4 | Independently, according to my own needs, and solving well-defined and non-routine problems, I can: | <ul style="list-style-type: none"> <li>illustrate information needs,</li> <li>organise the searches of data, information and content in digital environments,</li> <li>describe how to access these data, information and content, and navigate between them,</li> <li>organise personal search strategies.</li> </ul>                                                                           |
| ADVANCED           | 5 | As well as guiding others, I can:                                                                   | <ul style="list-style-type: none"> <li>respond to information needs,</li> <li>apply searches to obtain data, information and content in digital environments,</li> <li>show how to access these data, information and content and navigate between them,</li> <li>propose personal search strategies.</li> </ul>                                                                                 |
|                    | 6 | At advanced level, according to my own needs and those of others, and in complex contexts, I can:   | <ul style="list-style-type: none"> <li>assess information needs,</li> <li>adapt my searching strategy to find the most appropriate data, information and content in digital environments,</li> <li>explain how to access these most appropriate data, information and content and navigate among them,</li> <li>vary personal search strategies.</li> </ul>                                      |
| HIGHLY SPECIALISED | 7 | At highly specialised level, I can:                                                                 | <ul style="list-style-type: none"> <li>create solutions to complex problems with limited definition that are related to browsing, searching and filtering of data, information and digital content,</li> <li>integrate my knowledge to contribute to professional practice and knowledge and guide others in browsing, searching and filtering data, information and digital content.</li> </ul> |
|                    | 8 | At the most advanced and specialised level, I can:                                                  | <ul style="list-style-type: none"> <li>create solutions to solve complex problems with many interacting factors that are related to browsing, searching and filtering data, information and digital content.</li> <li>propose new ideas and processes to the field.</li> </ul>                                                                                                                   |



**Cierre**



# Bibliografía

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 **“Data Quality Fundamentals”**, B. Moses, L. Gavish, M. Vorwerck, 2022 <https://www.oreilly.com/library/view/data-quality-fundamentals/9781098112035/>